Experiment-20 : Performance evaluation of free-space optical (fso) link under atmospheric turbulence using optisystem

AIM:

To design and simulate a Free-Space Optical (FSO) communication system using OptiSystem and evaluate its performance under different atmospheric turbulence conditions.

OBJECTIVES:

1. Analyze the effect of range on received power, considering the combined contributions of geometric spreading loss and atmospheric attenuation.

2. Evaluate how the combined effects of increasing range and attenuation influence received power and bit error rate (BER).

3. Assess the extent to which increased transmit power can compensate for attenuation-induced signal loss.

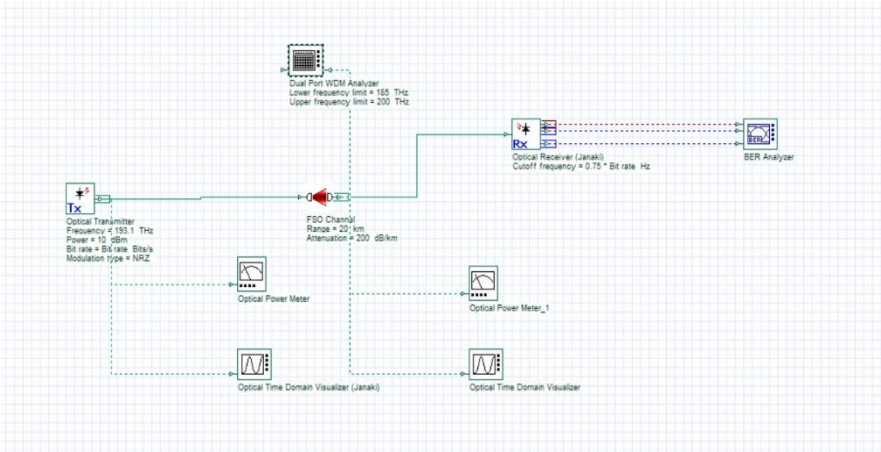
SOFTWARE REQUIRED:

• Optisystem

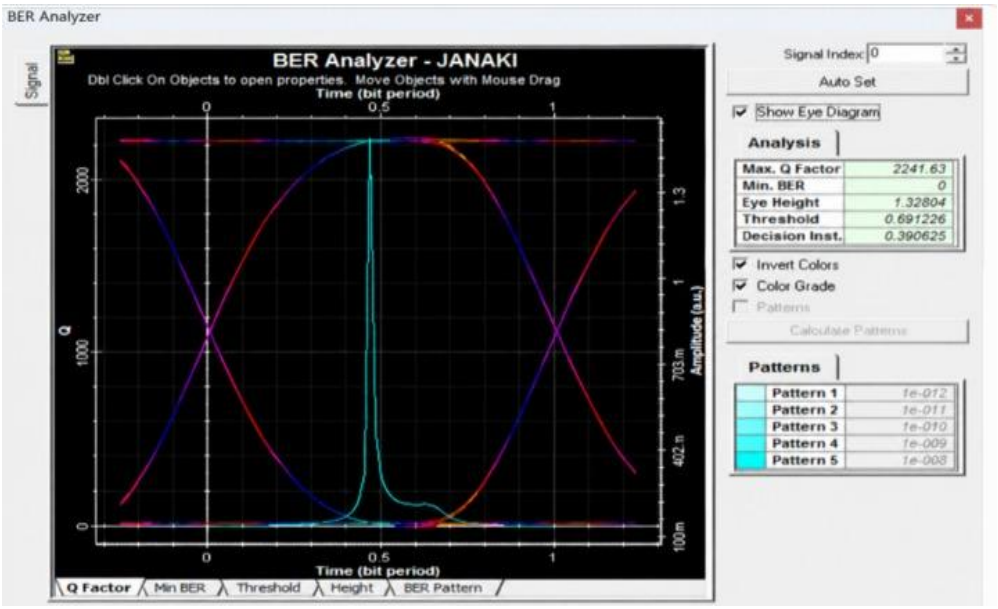
EXPERIMENT PROCEDURE:

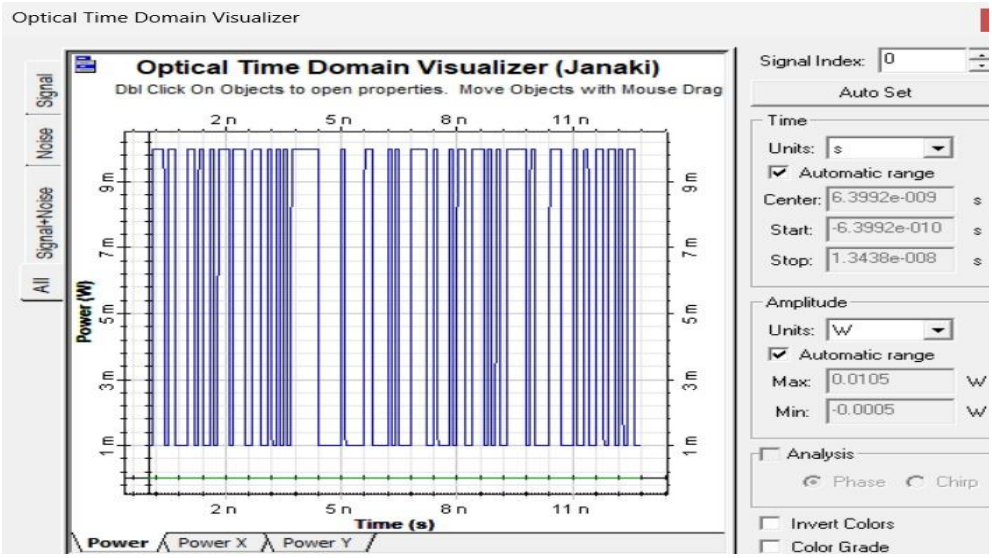
- Design the FSO transmi@er and receiver in OptiSystem using suitable optical source,modulator, photodetector, and receiver components.
- Set input parameters: bit rate, data pa@ern, optical power, and wavelength.
- Connect transmi@er, FSO channel, and receiver; configure transmission range and a@enuation values.
- Connect BER Analyzer, Eye Diagram Analyzer, and Optical Power Meter to monitor
- performance.
- Vary range and a@enuation, and simulate each condition.
- Record BER, Q-factor, eye diagram, and received power for each case.
- Compare results to analyze the effect of range and a@enuation on FSO link
- performance.

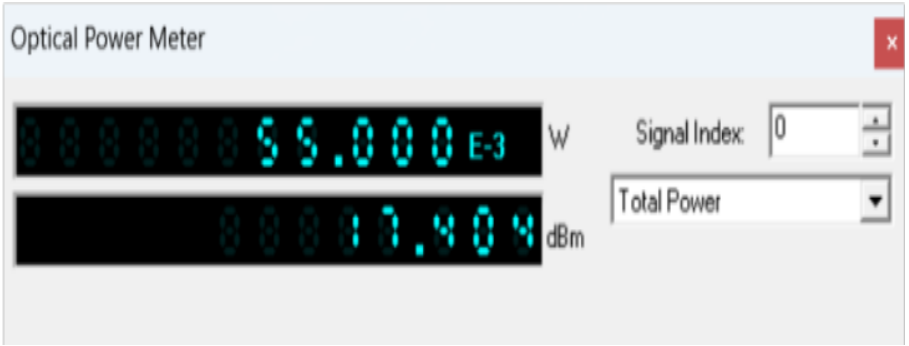
BLOCK DIAGRAM:

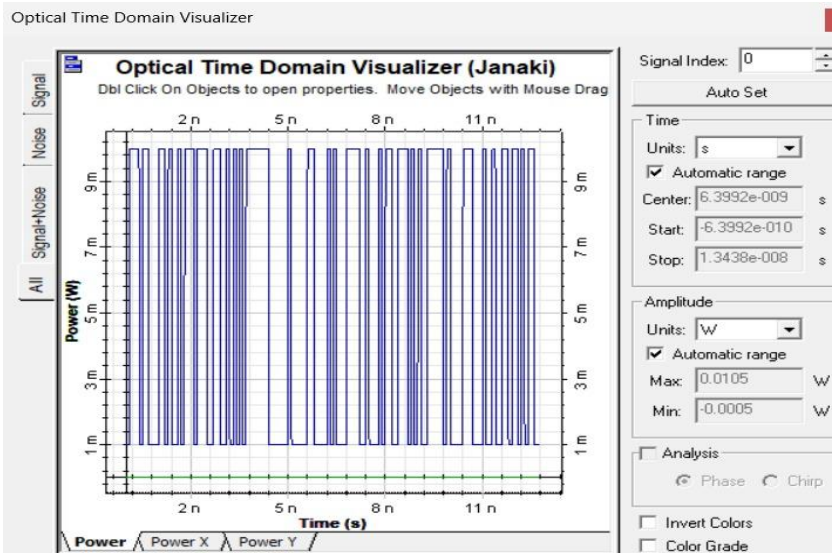


OUTPUT WAVEFORMS:











REFLECTION:

This project helped to understand the working of FSO communication and the effects of atmospheric turbulence on signal transmission. It improved knowledge of OptiSystem simulation, BER analysis, Q-factor measurement and optical communication system performance.

RUBRICS:

Assessment Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weightage
System Design	Correct FSO link design in OptiSystem.	Minor connection errors.	Partial system setup.	Incorrect setup.	20%
Parameter Configuration	All parameters correctly configured.	Minor parameter errors.	Some parameters missing.	Incorrect configuration.	20%
Measurement & Monitoring	Accurate power, BER and Q-factor readings.	Minor missing readings.	Partial measurements.	No proper measurements.	20%
Loss Analysis	Correct analysis of distance and atmospheric losses.	Basic loss analysis.	Limited analysis.	No proper analysis.	20%
Report & Conclusion	Complete report with graphs and results.	Good report and conclusion.	Incomplete report.	Poor documentation.	20%

RESULT:

The FSO link was successfully simulated using OptiSystem, and its performance was analysed under different atmospheric turbulence conditions.